Introduction to Loops in Python

- - Loops are used for iterating over a sequence.
- - Two main types of loops in Python:
  - `for` loop
  - `while` loop
- - Loop control statements:
  - `break`
  - `continue`
  - `pass`

For Loops

- - Used to iterate over a sequence (list, tuple, string, etc.)
- - Syntax:

```
for variable in sequence:
    # code block
```

- - Example:

```
for i in range(5):
    print(i)
```

-Loop with Else

- Syntax

  for i in range(3): print(i)

  else:

  print("Loop completed")

- The `else` block executes when the loop ends normally.

While Loops

- - Repeats as long as a condition is true
- - Syntax:

```
while condition:
    # code block
```

- - Example:

```
x = 0
while x < 5:
    print(x)
    x += 1
```

Loop Control Statements

- - `break`: Exits the loop prematurely
- - `continue`: Skips the current iteration and continues with the next
- - `pass`: Does nothing, acts as a placeholder

Break Statement

- \- Used to exit a loop before it finishes all iterations
- \- Example:

```
for i in range(10):
    if i == 5:
        break
    print(i)
```

Continue Statement

- - Skips the rest of the code inside the loop for the current iteration
- - Example:

```
for i in range(10):
    if i % 2 == 0:
        continue
    print(i)
```

Pass Statement

- - Placeholder when a statement is syntactically required but no action is needed
- - Example:

```
for i in range(5):
    if i == 3:
        pass
    print(i)
```

Summary

- - `for` and `while` are primary loop types
- - Control statements (`break`, `continue`, `pass`) provide loop control
- - Use loops to efficiently process data collections